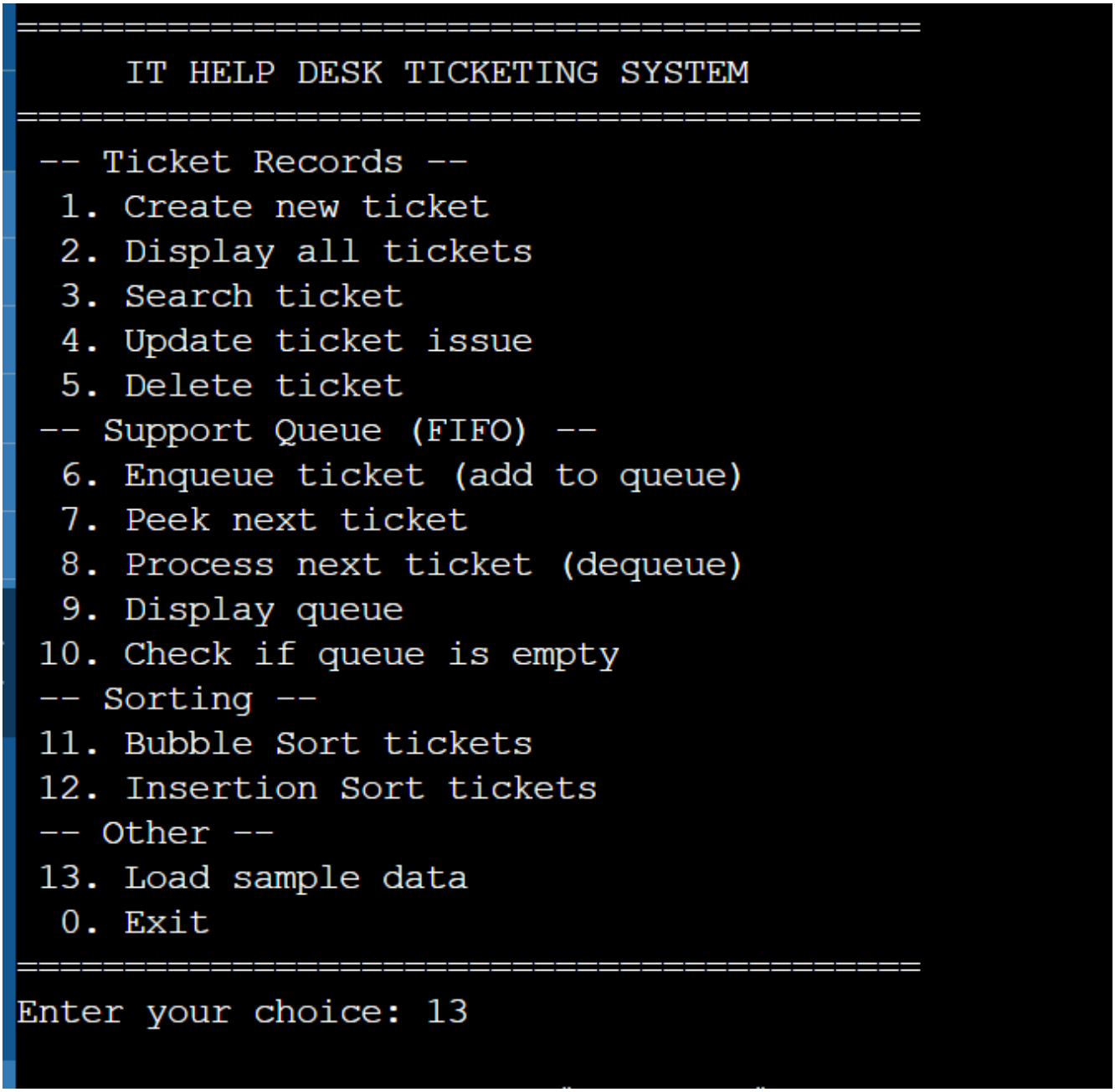
Screenshots Documentation

IT Help Desk Ticketing System - DSA Java Sorting and Queue Capstone

These screenshots were taken from the actual running program. Each one is explained below. The image files are in the **screenshots/** folder of the repository.

1. Main Menu

What it shows: the menu-driven interface when the program starts.



Main menu with all options (1-13 and 0 to exit). Option 13 was chosen to load sample data.

2. Adding and Displaying Records

What it shows: records are stored in ArrayLists, then displayed.

```
 11. Bubble Sort tickets
 12. Insertion Sort tickets
 -- Other --
 13. Load sample data
  0. Exit
==========================================
Enter your choice: 1

--- Create New Ticket ---
Requester name: Uri Jadwayne
Issue description: Campus wifi has no internet for me
Date submitted (yyyy-MM-dd): 2026-10-30
Ticket #1006 created successfully. (Status: Registered)
Tip: use option 6 to add it to the support queue.

==========================================
     IT HELP DESK TICKETING SYSTEM
==========================================
 -- Ticket Records --
  1. Create new ticket
  2. Display all tickets
  3. Search ticket
  4. Update ticket issue
  5. Delete ticket
 -- Support Queue (FIFO) --
  6. Enqueue ticket (add to queue)
  7. Peek next ticket
  8. Process next ticket (dequeue)
  9. Display queue
```

Option 1 (Create new ticket): the program asks for name, issue, and date, then creates ticket #1006.

```
  9. Display queue
 10. Check if queue is empty
 -- Sorting --
 11. Bubble Sort tickets
 12. Insertion Sort tickets
 -- Other --
 13. Load sample data
  0. Exit
========================================
Enter your choice: 2

--- All Tickets ---
Ticket#  Requester          Issue                          Date         Status
-------------------------------------------------------------------------------
1001     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1002     Juan Dela Cruz     Forgot student portal password 2026-10-03   Registered
1003     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1004     Carlo Mendoza      Printer jam in library         2026-10-01   Registered
1005     Bea Villanueva     Email account locked           2026-10-07   Registered
1006     Uri Jadwayne       Campus wifi has no internet for me 2026-10-30   Registered
Total tickets: 6

========================================
     IT HELP DESK TICKETING SYSTEM
========================================
 -- Ticket Records --
  1. Create new ticket
  2. Display all tickets
  3. Search ticket
```

Option 2 (Display all tickets): the 5 sample tickets plus the new ticket #1006 (Total: 6).

3. Queue Operations

What it shows: enqueue, peek, display, and dequeue working on the support queue.

```
  3. Search ticket
  4. Update ticket issue
  5. Delete ticket
 -- Support Queue (FIFO) --
  6. Enqueue ticket (add to queue)
  7. Peek next ticket
  8. Process next ticket (dequeue)
  9. Display queue
 10. Check if queue is empty
 -- Sorting --
 11. Bubble Sort tickets
 12. Insertion Sort tickets
 -- Other --
 13. Load sample data
  0. Exit
==========================================
Enter your choice: 6

--- Enqueue Ticket ---
Enter ticket number to add to the queue: 1003
Ticket #1003 added to the support queue. Position: 1

==========================================
     IT HELP DESK TICKETING SYSTEM
==========================================
 -- Ticket Records --
  1. Create new ticket
  2. Display all tickets
  3. Search ticket
```

Enqueue: ticket #1003 is added to the queue at position 1.

```
 3. Search ticket
 4. Update ticket issue
 5. Delete ticket
-- Support Queue (FIFO) --
 6. Enqueue ticket (add to queue)
 7. Peek next ticket
 8. Process next ticket (dequeue)
 9. Display queue
10. Check if queue is empty
-- Sorting --
11. Bubble Sort tickets
12. Insertion Sort tickets
-- Other --
13. Load sample data
 0. Exit
==========================================
Enter your choice: 6

--- Enqueue Ticket ---
Enter ticket number to add to the queue: 1001
Ticket #1001 added to the support queue. Position: 2

==========================================
     IT HELP DESK TICKETING SYSTEM
==========================================
-- Ticket Records --
 1. Create new ticket
 2. Display all tickets
 3. Search ticket
```

Enqueue: ticket #1001 is added at position 2.

```
  3. Search ticket
  4. Update ticket issue
  5. Delete ticket
 -- Support Queue (FIFO) --
  6. Enqueue ticket (add to queue)
  7. Peek next ticket
  8. Process next ticket (dequeue)
  9. Display queue
 10. Check if queue is empty
 -- Sorting --
 11. Bubble Sort tickets
 12. Insertion Sort tickets
 -- Other --
 13. Load sample data
  0. Exit
========================================
Enter your choice: 7

--- Peek Next Ticket ---
Next ticket to be processed (not removed):
Ticket#  Requester          Issue                          Date         Status
------------------------------------------------------------------------------
1003     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Queued
```

Peek: shows the next ticket (#1003, status Queued) without removing it.

```
Enter your choice: 9

--- Current Support Queue (front -> back) ---
1. Ticket #1003 | Ana Reyes | Projector not detected in Lab 2
2. Ticket #1001 | Maria Santos | Cannot connect to campus Wi-Fi
Tickets waiting: 2
```

Display queue: #1003 is at the front and #1001 is behind it (2 waiting).

```
Enter your choice: 8

--- Process Next Ticket ---
Now processing ticket #1003 from Ana Reyes.
Issue: Projector not detected in Lab 2
Ticket #1003 marked as Resolved.
Tickets still waiting: 1
```

Dequeue (process next): ticket #1003 is processed and marked Resolved. 1 ticket still waiting.

```
Enter your choice: 9

--- Current Support Queue (front -> back) ---
1. Ticket #1001 | Maria Santos | Cannot connect to campus Wi-Fi
Tickets waiting: 1
```

Display queue after dequeue: only ticket #1001 remains.

4. FIFO Behavior

What it shows: tickets are enqueued in the order 1004, 1002, 1005 (not numerical order) and are processed in that same order. First in, first out.

```
Enter your choice: 6

--- Enqueue Ticket ---
Enter ticket number to add to the queue: 1004
Ticket #1004 added to the support queue. Position: 1
```

Step 1: ticket #1004 is enqueued first (position 1).

```
----------------------------------------
Enter your choice: 6

--- Enqueue Ticket ---
Enter ticket number to add to the queue: 1002
Ticket #1002 added to the support queue. Position: 2
```

Step 2: ticket #1002 is enqueued second.

```
Enter your choice: 6

--- Enqueue Ticket ---
Enter ticket number to add to the queue: 1005
Ticket #1005 added to the support queue. Position: 3
```

Step 3: ticket #1005 is enqueued third.

```
Enter your choice: 9

--- Current Support Queue (front -> back) ---
1. Ticket #1004 | Carlo Mendoza | Printer jam in library
2. Ticket #1002 | Juan Dela Cruz | Forgot student portal password
3. Ticket #1005 | Bea Villanueva | Email account locked
Tickets waiting: 3
```

The queue lists the tickets in order of arrival: 1004, 1002, 1005.

```
Enter your choice: 8

--- Process Next Ticket ---
Now processing ticket #1004 from Carlo Mendoza.
Issue: Printer jam in library
Ticket #1004 marked as Resolved.
Tickets still waiting: 2
```

First process: ticket #1004 (the first one added) is processed first. 2 still waiting.

```
Enter your choice: 8

--- Process Next Ticket ---
Now processing ticket #1002 from Juan Dela Cruz.
Issue: Forgot student portal password
Ticket #1002 marked as Resolved.
Tickets still waiting: 1
```

Second process: ticket #1002. 1 still waiting.

```
Enter your choice: 8

--- Process Next Ticket ---
Now processing ticket #1005 from Bea Villanueva.
Issue: Email account locked
Ticket #1005 marked as Resolved.
Tickets still waiting: 0
```

Third process: ticket #1005. 0 still waiting.

```
Enter your choice: 9

--- Current Support Queue (front -> back) ---
The queue is empty.
```

Display queue: the program reports that the queue is empty.

5. Sorting Algorithm 1: Bubble Sort

What it shows: Bubble Sort (written manually) sorting the records by date submitted, oldest first. The queue order is not changed.

```
Enter your choice: 11

--- Bubble Sort ---
Sort by:
1. Ticket number
2. Requester name (A-Z)
3. Date submitted (oldest first)
Choose field: 3

BEFORE sorting:
Ticket#  Requester          Issue                          Date         Status
-------------------------------------------------------------------------------
1001     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1002     Juan Dela Cruz     Forgot student portal password 2026-10-03   Resolved
1003     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1004     Carlo Mendoza      Printer jam in library         2026-10-01   Resolved
1005     Bea Villanueva     Email account locked           2026-10-07   Resolved
1006     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1007     Juan Dela Cruz     Forgot student portal password 2026-10-03   Registered
1008     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1009     Carlo Mendoza      Printer jam in library         2026-10-01   Registered
1010     Bea Villanueva     Email account locked           2026-10-07   Registered
```

Records BEFORE Bubble Sort (sort field chosen: 3, date submitted).

```
AFTER Bubble Sort:
Ticket#  Requester          Issue                          Date         Status
-------------------------------------------------------------------------------
1004     Carlo Mendoza      Printer jam in library         2026-10-01   Resolved
1009     Carlo Mendoza      Printer jam in library         2026-10-01   Registered
1002     Juan Dela Cruz     Forgot student portal password 2026-10-03   Resolved
1007     Juan Dela Cruz     Forgot student portal password 2026-10-03   Registered
1001     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1006     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1005     Bea Villanueva     Email account locked           2026-10-07   Resolved
1010     Bea Villanueva     Email account locked           2026-10-07   Registered
1003     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1008     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered

Note: the support queue order was NOT changed (still FIFO).
```

Records AFTER Bubble Sort: dates run from 2026-10-01 to 2026-10-09. Ticket numbers break ties.

6. Sorting Algorithm 2: Insertion Sort

What it shows: Insertion Sort (written manually) sorting the records by requester name, A to Z.

```
Enter your choice: 12

--- Insertion Sort ---
Sort by:
1. Ticket number
2. Requester name (A-Z)
3. Date submitted (oldest first)
Choose field: 2

BEFORE sorting:
Ticket#  Requester          Issue                          Date         Status
-------------------------------------------------------------------------------
1004     Carlo Mendoza      Printer jam in library         2026-10-01   Resolved
1009     Carlo Mendoza      Printer jam in library         2026-10-01   Registered
1002     Juan Dela Cruz     Forgot student portal password 2026-10-03   Resolved
1007     Juan Dela Cruz     Forgot student portal password 2026-10-03   Registered
1001     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1006     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1005     Bea Villanueva     Email account locked           2026-10-07   Resolved
1010     Bea Villanueva     Email account locked           2026-10-07   Registered
1003     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1008     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
```

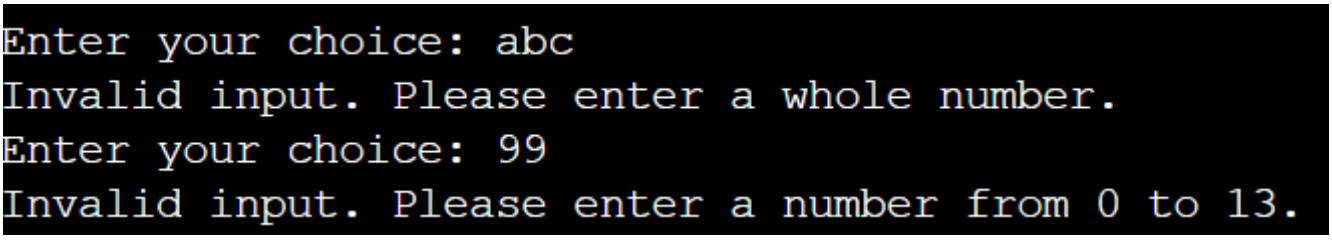
Records BEFORE Insertion Sort (sort field chosen: 2, requester name).

```
AFTER Insertion Sort:
Ticket#  Requester          Issue                          Date         Status
-------------------------------------------------------------------------------
1003     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1008     Ana Reyes          Projector not detected in Lab 2 2026-10-09   Registered
1005     Bea Villanueva     Email account locked           2026-10-07   Resolved
1010     Bea Villanueva     Email account locked           2026-10-07   Registered
1004     Carlo Mendoza      Printer jam in library         2026-10-01   Resolved
1009     Carlo Mendoza      Printer jam in library         2026-10-01   Registered
1002     Juan Dela Cruz     Forgot student portal password 2026-10-03   Resolved
1007     Juan Dela Cruz     Forgot student portal password 2026-10-03   Registered
1001     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
1006     Maria Santos       Cannot connect to campus Wi-Fi 2026-10-05   Registered
```

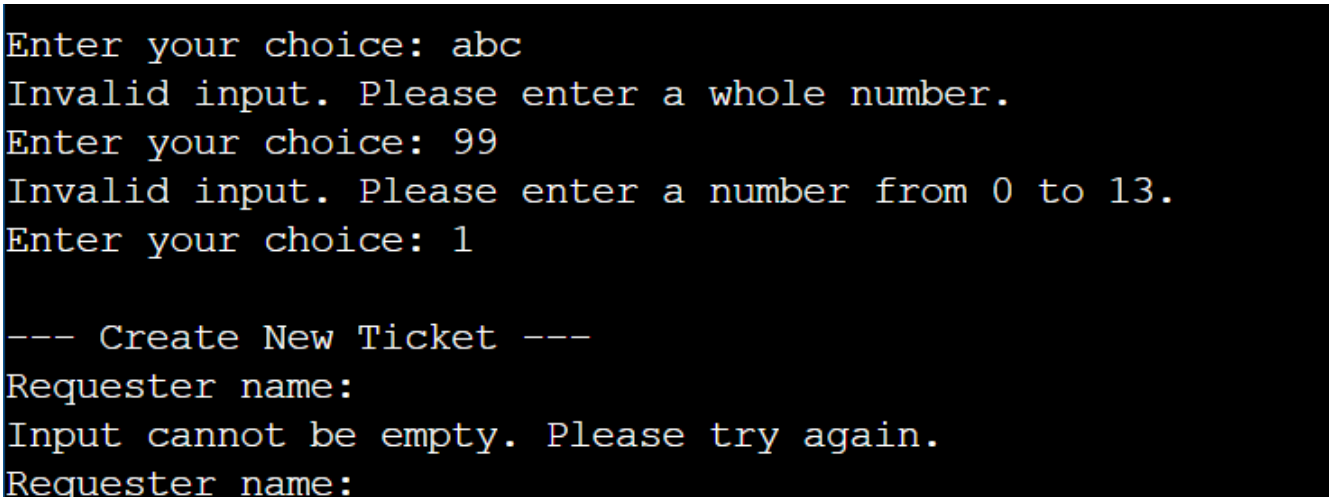
Records AFTER Insertion Sort: Ana, Bea, Carlo, Juan, Maria.

7. Input Validation

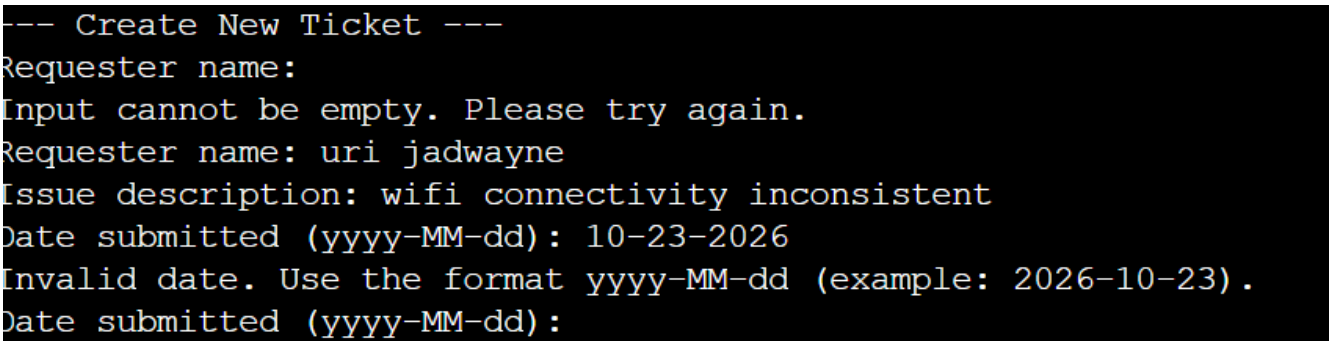
What it shows: invalid input is rejected with clear messages and the program keeps running.



Menu: typing letters (abc) and an out-of-range number (99) both show error messages.



Create ticket: pressing Enter on an empty requester name is rejected and the question is asked again.



Create ticket: a wrong date format (10-23-2026) is rejected and the program asks again.